Econ 494 Introduction to Stata

Instructor: Hyun Soo Suh

Spring 2024

E-mail: hyunsoo@wustl.edu
Office Hours: Fri 10:00A-12:00P
Office: Seigle 378
Software: Stata Statistical Software: Releases 16-18

Course Webpage: Canvas
Class Hours: Mon/Wed 5:30P-7:00P
Class Room: Seigle L016

Office Hours

Office hours will be held on Fridays between 10am and 12pm in my office (Seigle 378). I can also be reached via email, where I respond normally within 24 hours. However, please allow up to 48 hours before resending me an email. In case the scheduled office hours do not work for you, please feel free to email me, and we can arrange an alternative time, either in-person or online. I will be available for questions after each class.

Course Description

This is a half-semester course designed to familiarize students with advanced analytical tools using the statistical software Stata. The course aims to bridge the gap between econometrics and data analysis by utilizing real-world datasets. Regardless of your previous exposure to Stata, this course provides an opportunity to refresh and build upon your econometric knowledge and programming skills. This course teaches students to use advanced programming skills essential for any economics-related profession and graduate school.

The course is designed as a hands-on, learning-by-doing approach, with classes focused on lab sessions. Students will actively engage in tasks related to data preparation, analysis, and modeling, with a strong emphasis on real-life application in Economics.

Pre/Co-requisites

Students must have completed or be concurrently taking Econ 413: Introduction to Econometrics or equivalent.

Learning Goals

Upon completion of the course, students should be able to:

- Understand the econometric meaning of the commands
- Find, obtain, and prepare relevant databases for a research project
- Create data codebook and appropriate documentation for a research project
- Conduct descriptive statistics using survey data
- Estimate simple regression models and interpret results

Drop/Add Policy & Grade Options

This course must be taken as a letter grade. Options do NOT include P/F and audit. After the first session, students cannot use Webstac to add, drop, or withdraw from this course. The only exceptions are made in case of illness or emergency and all requests for changes in the grade option must be directed to Dorothy Peterson (dottie@wustl.edu).

Grading Policy

The course grade will be a weighted average of homework assignments, class labs, and the final project presentation. Details on the weights are described below. I expect the median grade to be above "B" and your progress will be updated through the Canvas course webpage. If you have any questions about your grade, please contact me.

1. **Homework Assignments (39%)**: There will be three homework assignments, consisting of reviews and applications using the tools taught in class. Please make sure to submit a good quality homework on time (e.g. do-file that runs properly with well explained comments detailing the commands, graphs with labels). The assignments will be graded based on the completeness and your demonstrated effort. A partial submission is acceptable and better than no submitted assignment because the latter will be graded as zero. In case you encounter a persistent issue while you are solving your homework (after multiple attempts on your part), you can reach me out with questions.

2. **Class Labs/Participation (30%)**: Given the course is held in a computer lab, class attendance is mandatory. During class, we will go through the lab lecture notes with exercises together. Before the following class, you will be required to submit your class lab work on Canvas. Since we have classes on Mondays and Wednesdays, the due dates will be Tuesdays and Sundays at 11:59pm. Please make sure you submit them on time.
   Lab work can be done in groups but should be submitted individually, meaning that you can discuss and solve problems together but should describe the work in your own words. Identical assignments between students will be penalized.
   Since participation is important, lab work submission and participation will be given equal weights. *In case of any emergencies, please let me know as soon as possible and we can come up with a solution.*

3. **Final Project Presentation (31%)**: For the final project, you will apply the tools learned in class to answer a research question of your interest using an appropriate dataset. You will be required to make a 15-20 minutes presentation (10-12 slides) explaining your motivation, research question, dataset, and findings. I will provide more details towards the end of the course. I will also schedule personalized meetings with each student one week before the presentation to discuss about your research project progress.

4. **Extra Credit (2.5%)**: You can get extra credit if you complete the course evaluation before the university deadline. I will give you a reminder through Canvas.

I will put strict deadlines on homework assignments. There will be no extensions. If you miss any deadline, you will receive an immediate zero for that assignment.

Based on the course grades, I will decide if it is necessary to impose a curve. However, I do not expect this to be the case.

Academic Integrity

All class lab work and homework should be your own. If you use graphs, references, or any materials from a website, book, or someone else, please make sure you cite them properly. Failing of citation can be subject to plagiarism and a failing grade. You will also be faced with strict and immediate academic disciplinary action.

Again, you are allowed to discuss and work with your classmates; however, you need to write and submit individually. Identical assignments will be penalized and will be considered as a violation of academic integrity.

For more details, please refer to the university's website on academic integrity policies. Lack of knowledge of the policy is not a legitimate excuse for violation.

Textbook

The main textbook is: *Colin & Trivedi (2010). Microeconometrics using Stata: Revised Edition, 2e. Stata Press, StataCorp LP: College Station, TX.*

Students are recommended to use the book as a supplementary tool; however, purchasing the book is not mandatory. I will provide self-sufficient lecture notes with detailed documents and exercises for students to revise the course materials without the textbook. Lecture notes will be shared before or after each class.

Statistical Software

For our class, access to Stata is mandatory. There are several ways to get Stata:

1. Purchase through Washington University's Gradplan for one semester or longer through this link. Version BE (for mid-sized datasets) or SE (for larger datasets) will be sufficient for the course.

2. Use the Arts & Sciences Computing Lab in Seigle Hall L016. I encourage students to use the computer labs during lab hours, which can be found in this link.

3. Use the **Apporto** cloud desktop. Login to the Apporto website, click 'Launch' under 'ArtSci' WUSTL Streamed App. This will give access to a virtual computer with Stata installed. For more details, please follow the guide at the Help Center.

Economic Datasets

Integrated Pulic Use Microdata Series (IPUMS), Current Population Survey (CPS), National Survey on Drug Use and Health (NSDUH), Medical Expenditure Panel Survey (MEPS).

Important dates

- **January 17th**: First day of class
- **February 5th**: HW1 deadline
- **February 14th**: HW2 deadline
- **February 26th**: HW3 deadline
- **March 6th**: Final project presentation/Last day of class

Tentative Course Outline

Lab 0-2: Introduction to Stata

Dataset: IPUMS-Current Population Survey (CPS)

Topics: Organize, do-file, log file, import, export, summarize, describe, tabulate, r-class, e-class, scalars, matrices, loops, global/local macros, label, generate, recode, drop.

Lab 3-6: Clean datasets and begin data analysis

Dataset: National Survey on Drug Use and Health (NSDUH)

Topics: preserve, restore, reshape, scatterplot, box-and-whisker plot, histogram, label graphs, graph export, merge, append, correlation, OLS regression.

Lab 7-9: Continue data analysis and conduct tests

Dataset: Medical Expenditure Panel Survey (MEPS)

Topics: hypothesis tests (Wald test, joint test, heteroskedasticity test, ommitted variables), prediction, interactions, survey weights, binary outcome models: Linear probability model (LPM), logit, probit, marginal effects.

Tentative Course Schedule

This is a tentative schedule, subject to change. If any changes are made, I will announce them on Canvas, so please make sure you check Canvas before each class.

The task refers to class lab work which is due normally the day before the following class. Most work will be done during the class, so the task will not be time consuming.

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday
			Jan 17 Lecture 0	18	19 Office Hours
21 Task 0 due	22 Lecture 1	23 Task 1 due	24 Lecture 2 HW1 distributed	25	26 Office Hours
28 Task 2 due	29 Lecture 3	30 Task 3 due	31 Lecture 4	Feb 1	2 Office Hours
4 Task 4 due	5 Lecture 5 HW1 due HW2 distributed	6 Task 5 due	7 Lecture 6	8	9 Office Hours
11 Task 6 due	12 Lecture 7	13 Task 7 due	14 Lecture 8 HW2 due HW3 distributed	15	16 Office Hours
18 Task 8 due	19 Lecture 9	20 Task 9 due	21 Lecture 10	22	23 Office Hours
25 Task 10 due	26 Lecture 11 HW3 due	27 Task 11 due	28 Lecture 12	29	Mar 1 Office Hours
3	4 **Open Lab**	5	6 **Final Project**	7	8